

Homework 1

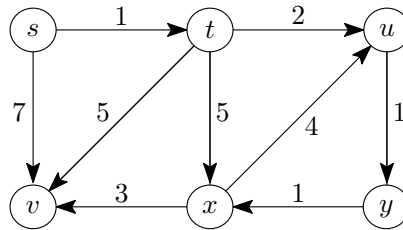
Graph Algorithms

UP FAMNIT, 2025/26

November 6, 2025

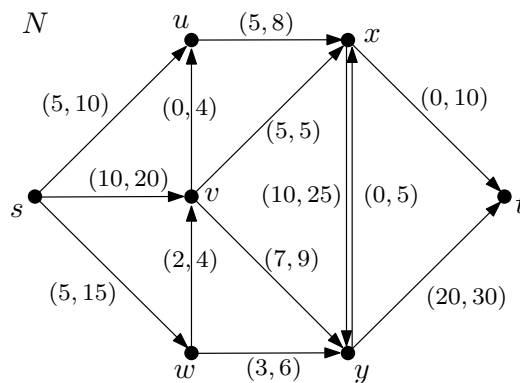
Maximum possible number of points: 50. Submission deadline: **Monday 17 November 2025 at 11:00**. Homework solutions have to be submitted via the e-classroom in PDF format (one file of a maximum size of 100 MB). You can prepare solutions in LaTeX, in Word, or by hand. If you write the solutions by hand, please make sure that the submitted material is legible. In this case, to create a pdf file, I recommend using the Notebloc application (which is freely available at <https://notebloc-shop.com/pages/descarga-app-notebloc-libretas-digitales>). I would appreciate a submitted solution on paper at the lectures of 17 November 2025.

1. [10 p] Consider the following edge-weighted digraph $D = (V, E)$.



- (2 p) Is (s, t, u, y, x, v) a topological sort of D ? Justify your answer.
- (2 p) Does the digraph D admit a topological sort? Justify your answer.
- (6 p) Find a tree of shortest paths from vertex s to all other vertices in D . Use Dijkstra's algorithm and write down all the steps of the algorithm, including the checks whether the values of the function $c(\cdot)$ change or not.

2. [8 p] The figure below depicts a flow f in a network $N = (D = (V, E), c, s, t)$, with ordered pairs $(f(e), c(e))$ on the edges.



- (4 p) Specify four s, t -cuts in the network, each containing four edges. For each such cut $C = \delta^+(X)$, list its capacity $\text{cap}(C)$ and the net outflow along the cut, that is, the value of $f(\delta^+(X)) - f(\delta^-(X))$.
- (4 p) Draw the residual network for the above flow and list four augmenting paths from vertex s to vertex t along with the maximum possible value of the flow increase along each of these paths.

3. [10 p] For a positive integer k , a digraph $D = (V, E)$ is **k -connected** if $|V| > k$ and for every two distinct vertices $u, v \in V$ such that $(u, v) \notin E$ there exist k internally vertex-disjoint (directed) u, v -paths. Consider the following decision problem:

DIGRAPH CONNECTIVITY

Input: A digraph $D = (V, E)$ and a positive integer k .

Question: Is D k -connected?

- (a) (4 p) Show that DIGRAPH CONNECTIVITY belongs to **NP** by describing a suitable certificate and explaining how to use it.
- (b) (6 p) Describe a polynomial-time algorithm for solving DIGRAPH CONNECTIVITY. Analyze the time complexity of the algorithm.

4. [6 p] Let G be an arbitrary graph and let M^* be a matching in the graph G and C^* a vertex cover such that $|C^*| = |M^*|$. Are the following statements correct or incorrect? Justify your answers.

- (a) (3 p) The matching M^* is a maximum matching in the graph G , and the vertex cover C^* is a minimum vertex cover.
- (b) (3 p) For each edge e of the matching M^* , the set C^* contains exactly one endpoint of e .

5. [16 p] Recall that the **NP**-complete INDEPENDENT SET problem takes as input a graph G and an integer k and asks whether G contains an independent set of size at least k . The **NP**-completeness proof shown in class can be done from 3-SATISFIABILITY, implying that INDEPENDENT SET remains **NP**-complete for input instances $(G = (V, E), k)$ that additionally satisfy $k \leq |V|/3$. Let us refer to this problem as ONE-THIRD-INDEPENDENT SET.

- (a) (3 p) Let $(G = (V, E), k)$ be an input to ONE-THIRD-INDEPENDENT SET, let $p = |V| - 2k$, and let $G' = (V', E)$ be the graph obtained from G by adding to it a set V_0 of p isolated vertices. Show that G has an independent set of size at least k if and only if G' has an independent set of size at least $|V'|/2$.
- (b) (5 p) Consider the following decision problem.

HALF-INDEPENDENT SET

Input: A graph $G = (V, E)$.

Question: Is there an independent set S in G such that $|S| \geq |V|/2$?

Using the result from (a), show that HALF-INDEPENDENT SET is **NP**-complete.

- (c) (3 p) A matching M in a graph $G = (V, E)$ is **perfect** if every vertex of G is an endpoint of some edge in M . Let G be a graph with a perfect matching M and let S be an independent set in G . Show that $|S| \geq |V|/2$ if and only if S contains exactly one vertex from each edge in M .
- (d) (5 p) Consider now the HALF-INDEPENDENT SET problem restricted to input graphs G that have a perfect matching. Show that in this case, the problem is solvable in polynomial time, by computing a perfect matching M in G and reducing the problem to an instance of 2-SATISFIABILITY. More precisely, by (c), the HALF-INDEPENDENT SET problem given G is equivalent to determining if G has an independent set S that contains exactly one vertex from each edge in M ; show that this latter problem can be reduced to 2-SATISFIABILITY.

Hint: for each edge $e \in M$, introduce one variable x_e , associating the two endpoints of e in an arbitrary way to the two literals x_e and $\overline{x_e}$.